

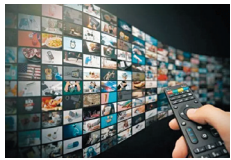
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

The Smallest GPUs for Seamless Visual Experiences

Imagination Technologies has launched a range of CXM Graphics Processing Units (GPUs). The CXM-2-64 is the smallest GPU in the range, offering



native support for high dynamic range (HDR) applications suitable for wearable devices,

smart home hubs, and mainstream set-top boxes. The CXM-4-64 has an efficient core designed for mainstream digital televisions (DTVs), and the CXM-4-128 offers a high-performance option for premium DTVs. Each GPU in the series boasts a nearly 50% performance density increase compared to the previous GPU family. The tiny frame buffer compression version 2 (TFBCv2) technology offers improved lossy/lossless compression, ensuring higher-quality compression and an additional compression level (37.5%) for enhanced design flexibility.

Imagination Technologies
<https://www.imaginationtech.com>

AI-Optimised Processor for Data-Heavy Storage Tasks

Kalray has launched Coolidge2, a new version of its 3rd generation Coolidge data processing unit (DPU) processor, known for its patented massively parallel processor array (MPPA) architecture. The processor is designed for elevated performance in artificial intelligence (AI) and general data processing. The high-speed DPU enhances the performance of large language model solu-

tions such as ChatGPT. It offers unique capabilities and precise control over neural network processing, including



memory transfers and utilisation of computing capabilities. The DPU delivers a significant performance boost, up to 10

times higher than the previous version. It offers a remarkable blend of computing power, energy efficiency, programming simplicity, high-speed interfaces, parallel application execution, deterministic operation, and real-time data processing.

Kalray
<https://www.kalrayinc.com>

High-Performance PMIC for Self-Powered IoT Devices

Nexperia has launched NEH2000BY, a high-performance power management integrated circuit (PMIC), to simplify and enhance the performance of low-



power applications, including Internet of Things (IoT) devices and embedded systems. The PMIC can efficiently charge batteries or storage capacitors by harnessing energy from ambient sources

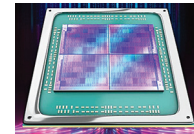
like light. The PMIC simplifies energy harvesting design and is up to 20 times smaller than competitors, with no manual optimisation needed. By eliminating inductors, the PMIC simplifies printed circuit board (PCB) design and substantially reduces the overall bill of materials (BOM) and board size, all within a compact 12mm² assembly area. Its unique algorithm quickly

adapts to environmental changes, maximising energy harvest.

Nexperia
<https://www.nexperia.com>

World's Largest FPGA-Based Adaptive SoC

The AMD Versal Premium VP1902 system-on-chip (SoC) has been recognised as the largest adaptive SoC globally. The



SoC is specifically designed for efficient verification of intricate semiconductor designs. With double the capacity com-

pared to its predecessor, this advanced solution empowers designers to drive innovation and validate application-specific integrated circuits (ASICs) and SoC designs. The SoC includes features such as automated design closure assistance, interactive design tuning, remote multi-user real-time debugging, and improved back-end compilation, which enables end users to iterate IC designs faster.

AMD
<https://ir.amd.com>

Industry's Top Multi-Zone Sensor for Enhanced Awareness

STMicroelectronics has launched a FlightSense multi-zone distance sensor with a 90° field of view, surpassing the previous generation by 33% and becoming the industry's top performer. The optical sensors enhance situational awareness in applications such as home



automation, domestic appliances, computers, robots, and smart equipment in stores, facto-